

ShopSphere - Customer Conversion & Marketing Effectiveness Analysis

Team Members: Anju Raveendran, Ayana S, Nikunj Dhanani

1. Business Problem Understanding and Objective Formulation

ShopSphere is a fast-growing e-commerce marketplace (electronics, fashion, home appliances, beauty, lifestyle) that has aggressively scaled digital marketing across search, social media, influencers, display, email, and affiliate networks, driving strong traffic growth. However, only a small fraction of these visitors completes a purchase, which inflates Customer Acquisition Cost (the spend to gain one customer) and erodes Return on Ad Spend. The core gap is visibility: existing reports show what happened at a high level, but not why particular customers or groups convert or abandon.

1.1 Key Business Problems

- **Low conversion** - high traffic but few completed purchases.
- **Marketing** - many clicks that do not turn into sales, pointing to mis-targeted spend.
- **Customer Experience** - inconsistent engagement across channels and touchpoints.
- **Product** - high cart abandonment and friction in the on-site journey.
- **Finance** - rising marketing budgets without matching revenue growth.
- **Insight gap** - aggregated reports lack the customer-level detail needed for targeting.

1.2 Business Objectives

- Optimise marketing spend and improve overall marketing effectiveness.
- Reduce customer acquisition cost and raise conversion rates through better targeting.
- Understand which channels, campaigns, and segments drive the most purchases.
- Support revenue growth and profitability through stronger conversion strategies.

1.3 Analytical Objectives

- Ingest and clean the historical customer interaction data.
- Explore customer behaviour, engagement, and campaign performance, including the funnel.
- Identify the factors that separate converting from non-converting visitors.
- Build a model to predict each visitor's likelihood of converting.
- Group customers into segments based on behaviour and engagement.
- Produce clear reports and practical recommendations for marketing decisions.

2. Challenges and Constraints

2.1 Challenges

- **Data quality and relevance** - legacy columns, a junk colour field, missing values, duplicates, inconsistent formats, and outliers must be cleaned, and redundant or weakly predictive columns dropped.
- **Missing channel cost** - spend per marketing channel is not recorded, which limits exact ROI and CAC calculation.
- **Data leakage** - fields such as discount, shipping, and cart value appear only after a purchase, so they must be excluded to avoid giving away the answer.
- **Rare buyers** - very few visitors buy, so accuracy alone is misleading, and we judge the model with metrics suited to rare events.
- **Granularity, behaviour and attribution** - data is mostly at session level, segments behave very differently, and multi-channel journeys make crediting one channel and proving cause difficult.
- **Explainability and adoption** - stakeholders moving from macro reports need clear, nonblack-box results, and live use needs engineering effort and real-world testing.

2.2 Assumptions

- The dataset accurately represents normal historical behaviour, with conversion clearly defined.
- Each record correctly links to a visit and its purchase outcome, and tracking is reasonably reliable.

2.3 Limitations

- Only the provided columns are used, so findings show patterns rather than proven causes and are bounded by data quality.
- Based only on ShopSphere's own data, so external factors such as seasonality, competitor activity, pricing, and market conditions may be missed.
- Without verified per-channel cost, budget and ROAS guidance is directional (relative), not exact.

2.4 Constraints

- Customer data must follow privacy regulations.
- Time and page limits keep the focus on analysis rather than a full live system, so recommendations may stay at strategy or prototype level.
- Only available tools are used, and the solution must stay easy to explain.

3. Background Study / Literature Review

3.1 Conversion and Cart Abandonment

Purchase intent is widely modelled using session and clickstream data, where behavioural signals are the strongest predictors. Cart abandonment is mainly driven by checkout friction, hidden costs, and other journey factors (Sakar et al., 2019; Kukar-Kinney & Close, 2010; Rajamma et al., 2009).

3.2 Customer Segmentation

RFM (Recency, Frequency, Monetary) analysis and clustering methods such as K-Means and hierarchical clustering group customers by behaviour and value, enabling personalised marketing.

3.3 Machine Learning and Interpretability

Logistic Regression gives an interpretable baseline, while Random Forest and XGBoost are widely used for accurate prediction. Imbalance is handled with SMOTE and evaluated using ROC-AUC and PR-AUC. SHAP explains key drivers, and leakage prevention ensures trustworthy models (Breiman, 2001; Chen & Guestrin, 2016; Chawla et al., 2002; Lundberg & Lee, 2017; Kaufman et al., 2012).

3.4 Research Gap

Most studies focus mainly on prediction accuracy. This project adds explanation, segmentation, and channel-level guidance on a single dataset while explicitly removing legacy and leakage attributes.

4. Proposed Methodology / Solution Approach

- 4.1 Data acquisition & understanding - collect customer interaction and marketing data, perform EDA, and assess data quality, distributions, and behaviour.
- 4.2 Data preprocessing - remove identifiers, legacy and junk columns and leakage fields, handle missing values and outliers, encode categorical variables and scale features, address imbalance with SMOTE or class weights.
- 4.3 Feature engineering – Generate RFM (Recency, Frequency, Monetary) features, behavioural and engagement metrics, and extract temporal and campaign features.
- 4.4 Segmentation & analysis - cluster customers, analyse conversion trends, identify funnel drop-offs and cart-abandonment patterns.
- 4.5 Predictive Model development – train classification models such as Logistic Regression, Random Forest, XGBoost, and LightGBM with hyperparameter tuning and cross-validation.
- 4.6 Model evaluation - assess performance primarily using PR-AUC and ROC-AUC, supported by precision, recall, and F1-score, and interpret predictions using feature importance and SHAP.
- 4.7 Insights & recommendations - identify key conversion and abandonment drivers, recommend campaign, segmentation, personalisation and retargeting actions with directional budget guidance.

5. Proposed Timeline

The project is planned across six phases, summarised below with the target completion and effort for each.

Phase	Complete by (effort)	Activities
Data Ingestion & Pre-processing	05/07/2026 (1 week)	Data audit, handling missing values and duplicates, standardising data types, and establishing master data for analytical use.
Exploratory Data Analysis & Funnel Diagnostics	12/07/2026 (1 week)	Descriptive statistics, multi-channel clickstream tracking, funnel drop-off calculations, and demographic behaviour analysis.
Behavioural Customer Segmentation	26/07/2026 (1-2 weeks)	Engineering behavioural variables, performing clustering and profiling, and mapping channel-specific engagement trends.
Feature Engineering & Machine Learning Predictive Modelling	16/08/2026 (3-4 weeks)	Identifying features, training classification algorithms, optimising hyperparameters, and evaluating key factors for final model selection.
Strategic Framework for Explainability	30/08/2026 (2 weeks)	Extracting global and local feature importance and translating insights into actionable strategies per functional team.
Final Reporting	06/09/2026 (1 week)	Compiling final technical documentation, designing executive slide decks for stakeholders, and drafting deployment roadmaps.

6. Expected Deliverables & Outputs

- **Data Preprocessing & Audit Report:** a comprehensive technical summary outlining missing-value remedies, outlier treatments, and transformations of legacy marketing variables.
- **Funnel & Segmentation Insights Profile:** an analytical breakdown detailing consumer personas, channel performance benchmarks, and explicit funnel-stage transition metrics.
- **Predictive Conversion Model Portfolio:** trained classification models, hyperparameter-validation assets, evaluation metrics, and model interpretability plots.
- **Data-Driven Strategic Recommendation Roadmap:** tailored guidelines - precision target profiles for Marketing, friction-removal points for Product, personalised cadences for CX, and a spend-reallocation framework for Finance.
- **Final Capstone Presentation & Documentation:** A polished executive deck and project report prepared.